

Although VLSI companies typically hire at M.Tech levels for VLSI design positions, candidates with good fundamentals in electronics, electrical or computer science engineering can find positions in areas such as testing and debugging of VLSI chips, informs Sen. He says, "Embedded software, which involves writing code for microcontrollers/processors embedded within VLSI chips, is also an active area of recruitment."

"At our company, we recruit fresher graduates for working in the VLSI domain, provided they are good at the fundamentals of digital logic design (also referred to as digital electronics/switching theory and logic design), digital system design and have done their project implementation in either VHDL/Verilog HDL. We also give internships to M.Tech (in VLSI) students for their second year project work. For internships, we conduct written exams, technical and HR interviews," shares Raju. He adds, "Currently, our team comprises more than ten engineers working in the field of VLSI. We are planning to recruit ten more for this year. Our recruitment procedure is always in internship mode."

Pay package and demand areas

"During internship, we pay a stipend of anywhere between ₹ 4000 and ₹ 8000 per month, depending on the performance in the internship recruitment procedure. After completion of the internship/project-work, we offer the candidate a job in our company. The salary emolument is typically around ₹ 144,000 to ₹ 300,000 annually, depending on individual capabilities in technology," informs Raju. Although, he maintains, MNCs pay double the amount of what they offer at Unistring.

As a skill, chip designers form the cornerstone for electrical and computer engineering domains, and graduates from good institutes can command handsome starting salaries. As per figures provided by TimesJobs.com, 55 per cent of the engineers (basically

Expert advice

"To get into FPGA based embedded or digital design companies, one must have strong fundamentals in digital logic design, digital system design and must be capable of writing proper VHDL/Verilog HDL code that correctly works on FPGA."

— **K. Srinivasa Raju, CEO, Unistring Tech Solutions Pvt Ltd**

"Students interested in pursuing a career in the VLSI industry must build a foundation in digital design and various VLSI design flows by gaining practical skills during their undergraduate studies."

— **Harish Mysore, director, India operations, Global IEEE Institute for Engineers Pvt Ltd**

"Take up this career only if you are ready for the long haul. Be patient. Try to become inter-disciplinary and acquire new skills. Small start-up companies pay those engineers well who have multiple skills."

— **Subhajit Sen, associate professor, International Institute of Information Technology, Bangalore**

"My main advice to people would be to work hard and focus on the basics. Once you lay a strong foundation for yourself, it is very easy to build on top of it. Look at the quality of the work and stay focused on it in the initial years, as this will help you become a very strong engineer later in your career."

— **Uday Ramachandran, vice president, Bangalore operations, Interra Systems India Pvt Ltd**

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feeshers) working in the VLSI domain are paid anywhere between ₹ 120,000 and ₹ 300,000 per annum. Twenty three per cent engineers draw anywhere between ₹ 400,000 and ₹ 700,000 per annum and 13 per cent of the (senior) engineers in this industry are paid nearly ₹ 1 million and above.

"The silicon valley of India, Bengaluru, accounts for a majority of the jobs, as most companies have their core technology centres based there, with Hyderabad, Delhi, Chennai and Pune accounting for other opportunities in the VLSI sector," informs Madhukar.

In the near future

The VLSI industry is expected to grow rapidly in the next few years due to further reduction in geometry,

reduced power requirements and very large-scale application-specific integrated circuits. As a result, continuous investments will be made by integrated device manufacturers in the semiconductor industry, in addition to the various steps taken by the Indian government to boost indigenous production of electronics in India to address rapid growth in the local demand for electronics. "Further, as a result of the increasing use of electronics in telecom, healthcare, automotive, industrial and office automation, and consumer goods, the need for advanced chip design and verification engineers in the semiconductor industry will further increase," notes Mysore. ●

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